

BEVTrack: Multi-View Multi-Human Registration and Tracking in the Bird's Eye View

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Abstract—We handle a new problem of multi-view multi-human tracking in the bird's eye view (BEV). Different from previous works, we require neither the calibration among the multi-view cameras nor the actually captured BEV video. This makes the studied problem closer to real-world applications, however, more challenging. For this purpose, in this work, we propose a novel BEVTrack scheme. Specifically, given multi-view videos, we first use a virtual BEV transform module to obtain the BEV for each view. Then, we propose a unified BEV alignment module to fuse the respectively generated BEVs, in which we specifically design the self-supervised losses by considering both the spatial consistency and the temporal continuity. During the inference, we design the camera-subject collaborative registration and tracking strategy to make use of the mutual dependence between the multi-view cameras and the multiple targets, to achieve the desired BEV tracking. We also build a new benchmark for training and evaluation, the experimental results on which have verified the rationality of the problem and the effectiveness of our method.

Index Terms—BEV, human association, multi-human tracking, multi-view video, video surveillance, wearable cameras.

I. INTRODUCTION

MULTI-HUMAN detection and tracking is a fundamental and significant problem in the computer vision community. Previous works on human detection and tracking problems commonly use the single-view camera. Recently, a series of works have begun to use multi-view videos for collaborative tracking [1], [2], [3] to handle the problems of mutual occlusion and out-of-view, which are common in multi-human scenarios. More recently, the bird's eye view (BEV), also called the top view, has drawn the attention of researchers in the field of detection and tracking [4], [5]. We can imagine a scenario with

multiple moving humans in an outdoor environment (such as gatherings, sports), captured by multiple cameras. If we can obtain the global perspective from the BEV, including the clear spatial distribution and temporal trajectory, it is highly useful in various real-world applications, e.g., intelligent video surveillance, sport analysis systems, etc.

Some previous works have studied the human detection and/or tracking in the BEV. For instance, multi-view multi-human detection in a (virtual) BEV from a single image has been studied in [6], [7]. Additionally, a series of works [4], [5] use a top-view camera (on a drone) to obtain the real BEV video for collaborative analysis with the mobile cameras on the ground, which can provide a better human detection and tracking performance. Besides the perception of humans, BEV detection and tracking are prevalent in autonomous driving [8], [9], [10]. These works incorporate depth sensors like LiDAR/Radar to capture the point cloud data in the BEV [11], [12] or use multiple *calibrated* RGB cameras (usually six circularly arranged cameras) for collaborative detection and tracking [13], [14].

However, there are two main limitations existing in the above works. The first category of existing studies [6], [7], [13], [14] requires the camera calibration given in advance, which is demanding and not very applicable in many real-world scenes. For instance, the camera calibration may be unknown or have been changed, not to mention mobile cameras without fixed camera calibration. The second category of works uses special equipment, e.g., a drone to provide the real top-view videos [4], [5], or the LiDAR to obtain the point cloud map in the BEV [11], [12], [15]. However, using such additional equipment is not very practical. For example, maintaining a flying drone for a long term is not available in many real-world surveillance scenes.

Therefore, we aim to explore a question that **can we achieve the multi-human unified BEV tracking with only the free first-person-view (FPV) cameras** (freedom of movement and rotation without relative-pose calibration). From this question, in this work, we handle the problem of *multi-view multi-human tracking in the bird's eye view* (BEVTrack), without given multi-view camera calibration or additional equipment for BEV capturing. Specifically, as shown in Fig. 1, we aim to track multiple humans (referred to as subjects in the following) in a bird's eye view, utilizing multiple first-person-view (fixed/mobile) cameras without the given calibration. The proposed task *breaks the vital limitation* of camera calibration or BEV equipment, which is *significant for the multi-camera collaborative BEV perception*. The BEV fuses multiple first-person

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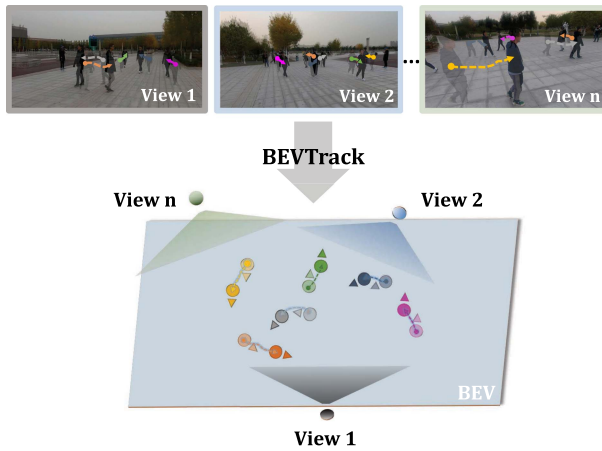


Fig. 1. An illustration of the BEVTrack, which takes multiple first-person-view (FPV) videos as input, and generates the unified tracking results in the BEV.

views (FPVs) to track the subjects, which can effectively handle the problem of mutual occlusion or subject missing in some FPV cameras. This work specifically focuses on the subject of humans, since it is specifically significant with many practical applications.

Although practical, this is also a particularly challenging problem since 1) Given the mobile cameras and moving targets, the camera relative relationships are inconstant over time and not easy to obtain, 2) The cross-view subject matching/fusion and over-time tracking in the BEV become particularly challenging given very large view differences. To achieve this, a straightforward idea for this problem may be the local descriptors-based methods for multi-view camera pose estimation [16], [17], [18]. However, these algorithms can not handle the proposed problem since they need the input images to have enough overlapped area with textural information. In practice, the overlap may be very limited given the large view difference (even on the opposite side), and the scene (dominated by humans) is unfavorable for local descriptor extraction.

In this work, we propose a novel solution to address the aforementioned challenges in the BEVTrack problem. Our basic insight is to *leverage the principle of spatial-relation dependence between the multiple cameras and the multiple subjects*. We observe that the subject distribution in the real 3D world is unified, which presents variously in the 2D image given different camera poses. This way, we first restore the subject 3D localization in the respective BEV from each camera. The next key problem is to integrate the BEVs from different cameras, and obtain the camera and subject detection (registration) results in a unified BEV. For this purpose, we make use of the prior of the unified subject spatial distribution in the 3D world to estimate the relative camera pose (i.e., from subject to camera), and based on the camera registration to further refine the subject registration in the BEV (i.e., from camera feedback to subject).

Based on this insight, we propose a joint framework to simultaneously achieve the subject/camera tracking (including spatial registration and temporal tracking) in the BEV. Specifically, in each FPV, we first apply a view-transform subject detection

module (VTM) to obtain the subject detection results in the BEV. Then, based on the above insight of spatial-relation dependence, we propose an imaging-geometry-based spatial alignment module (SAM) to estimate the relative pose of the multiple cameras in the BEV, in which we also develop a multi-view human matching strategy to obtain the cross-view human correspondence among the multiple views. After that, we specifically design a series of *self-supervised* losses considering both the spatial consistency and temporal continuity of the subjects, for the network training. Note that, the whole framework requires only the supervision of the camera registration but not the subject registration in BEV, which is a weak-supervised method. In the inference stage, with the camera pose estimation from SAM, we apply a camera pose selection strategy and a subject fusion scheme to get the camera and subject registration results in the unified BEV. Finally, we design a camera-subject-cooperative tracking strategy to achieve the BEVTrack. The main contributions of this work are:

- We propose to study a new and fundamental problem for multi-view free-camera unified BEV perception. To the best of our knowledge, this is the first work to achieve the camera and subject registration and tracking in BEV with multiple uncalibrated FPV cameras, which breaks the limitations of using pre-given camera calibration or real BEV images in previous works.
- We design a novel spatial alignment module (SAM) to leverage the multi-view spatial geometry relations and develop a series of self-supervised spatial-temporal consistency losses. They are integrated into a learnable framework, which possesses both the generalization of the deep network for the human localization and association task, and the explainability of classical geometry for the camera/subject pose alignment problem.
- We specifically design an inference framework for BEVTrack, which involves a camera-subject joint registration and tracking strategy by fully excavating the mutual dependence between the cameras and subjects for collaborative BEV registration and tracking.
- We provide the foundation of BEVTrack. We build a new benchmark composed of a large-scale synthetic video dataset and a re-constructed real-world evaluation dataset. Extensive experimental results on the synthetic dataset and the cross-dataset evaluation on the real-world dataset show the effectiveness of our method, and the rationality of the proposed BEVTrack problem.

Preliminary results of this work have been published in a conference proceeding called BEVSee [19]. Compared with the published conference paper, this paper is rewritten with many new contents and the following extensions. First, based on the *single-image-level* subject registration approach [19], we develop a new method for BEVTrack to achieve the temporal tracking for *videos* in BEV. Both the self-supervised spatial-temporal learning loss (Section III-C) and the camera-subject collaborative tracking (Section III-F) are newly proposed in this work. Second, we no longer use the previous dataset in [19] composed of static images, and build a larger new video benchmark used for BEVTrack problem (Section IV). Third, we re-implement the experiments for BEVTrack on this new benchmark (Section V).

The main results of the comparison methods and our method, and the ablation study on the benchmark are all newly presented in this paper. Finally, we have rewritten the introduction and related work sections and added the discussion section (Section VI) to make this paper more comprehensive.

II. RELATED WORK

Multi-view object detection aims to gather information from different cameras with various viewpoints, which can well enhance the detection performance, especially for occluded objects. Recent approaches for this task use pre-given camera calibration to map the object positions from the 2D image to real-world 3D coordinates and then generate a shared detection plane from the BEV. The approaches can be categorized into two primary types: the anchor-box based methods and the anchor-free based ones. For the anchor-box based approaches [6], [20], the first step is detecting the object's anchor box. After that, the features of the box and camera calibration are utilized to determine the object's position on the shared BEV plane. On the other hand, the anchor-free approaches [7], [21], [22] start by extracting the features of the entire image, followed by using the homography matrix (from the camera calibration) to project them onto a shared BEV plane without anchors. The above mentioned works all rely on camera calibration for cross-view projection, which limits their real-world applications. For example, they cannot be applied to the scenarios without accurate calibration parameters in prior, not to mention the mobile cameras without fixed calibration. A recent work [19] expands the multi-view human detection using mobile cameras without calibration. Compared with the work [19], which focuses on image-based detection, this work extends its scope to handle videos and achieves temporal tracking. For this purpose, we propose the spatial-temporal self-supervised losses and a cooperative tracking strategy, which are distinctly different from [19].

Multi-view multi-object tracking aims to utilize the multi-view videos to collaboratively track the objects in a scene [2], [4], which has the advantage of identifying the target even if it disappears (or is occluded) for a long time, by continuously tracking it using other views. Some previous works [1], [2], [23], [24], [25], [26], [27] focused on the multi-view multi-object tracking using cameras with overlapping field of view, where the objects can be captured by multiple views simultaneously. Most of them use fixed cameras for capturing videos. Recently, several advanced multi-view tracking methods have been proposed. ReST [28] employs a reconfigurable spatial-temporal graph model for multi-camera multi-object tracking. CrossMOT [29] uses decoupled multi-head embedding with conflict-free losses to jointly learn cross-view Re-ID features while resolving conflicts between cross-view and single-view embeddings. CRTracker [30] enhances cross-view association beyond CrossMOT by introducing a prediction module that leverages frame-to-frame association results as detections and fusion scores as confidences, strengthening temporal associations across multiple views. MvMHAT [3] performs multi-view tracking with several egocentric mobile cameras using a spatial-temporal association network with a self-supervised

learning method. Besides, some studies [4], [5] handle a new yet challenging problem of human association and tracking through both the bird's eye view from the drone and the side views from the egocentric wearable cameras. Nonetheless, these methods are restricted to a particular scenario where a drone captures the bird's-eye view video. This may hinder the applicability in the real world. In this work, we do not need a real BEV like [5]. We generate the BEV map, including the object tracking aggregated from all side views. Also, this work focuses on the scenario of multi-human *tracking in BEV* using the cameras *with FOV overlap*, which has a significant difference with previous tasks of multi-camera multi-target tracking without FOV overlap for people, e.g., DukeMTMCT [31], [32], or vehicle, e.g., AI City [33], [34]. Note that, these works only consider the object tracking in the input videos but not the BEV tracking. Compared with previous works, we can obtain the object tracking results in the generated BEV, but not require the actually captured BEV video, which is more suitable for the global observation of the whole scene but without additional equipment.

BEV perception in autonomous driving is another series of works related to this work. Based on the input data, they can be divided into two parts, i.e., BEV camera and BEV fusion [10]. BEV camera indicates using multiple surrounding vision cameras (fixed on the vehicle) with calibration for 3D object detection. BEV fusion depicts the fusion mechanisms from multiple sensor inputs, such as camera, LiDAR, and radar. Recent advances in multi-modal BEV fusion have shown significant improvements in 3D perception. BEVFusion [35] introduces a decoupled fusion framework that enables camera streams to operate independently from LiDAR data, addressing sensor failure scenarios. The framework extracts BEV features from both camera and LiDAR streams separately, then fuses them in the BEV space through simple concatenation. For radar-camera fusion, RCBEVDet [36] proposes a specialized radar feature extractor with RCS-aware BEV encoding and cross-attention multi-layer fusion to handle radar's sparse characteristics and alignment challenges. Its extension RCBEVDet++ [37] further enhances the framework with sparse fusion and support for query-based models. The modern 3D tracking methods [8], [9], [38], [39] are often based on the tracking-by-detection scheme, i.e., detecting objects in the current frame and linking them to previously tracks. Some works enhance their associations by incorporating learned motion and appearance features [40], [41], [42], or utilizing graph neural networks [43], [44], [45]. All of the works mentioned above are good at vehicle 3D tracking, but they all require a high-precision depth camera for localization, or multiple surrounding vision cameras with accurate calibration, which is not very economical and practical. Differently, our work mainly focuses on the human subjects. Moreover, it detects the subjects from the side views (through freely-moving cameras) and regresses their positions and trajectories into a BEV map, without the need for additional sensors, such as LiDAR, or the camera calibration.

BEV video analysis: Recently, some works have proposed to associate the top view (BEV) with the first-person views for collaborative analysis. The challenge VisDrone [46], [47] aims to provide a benchmark for the detection, single/multiple object

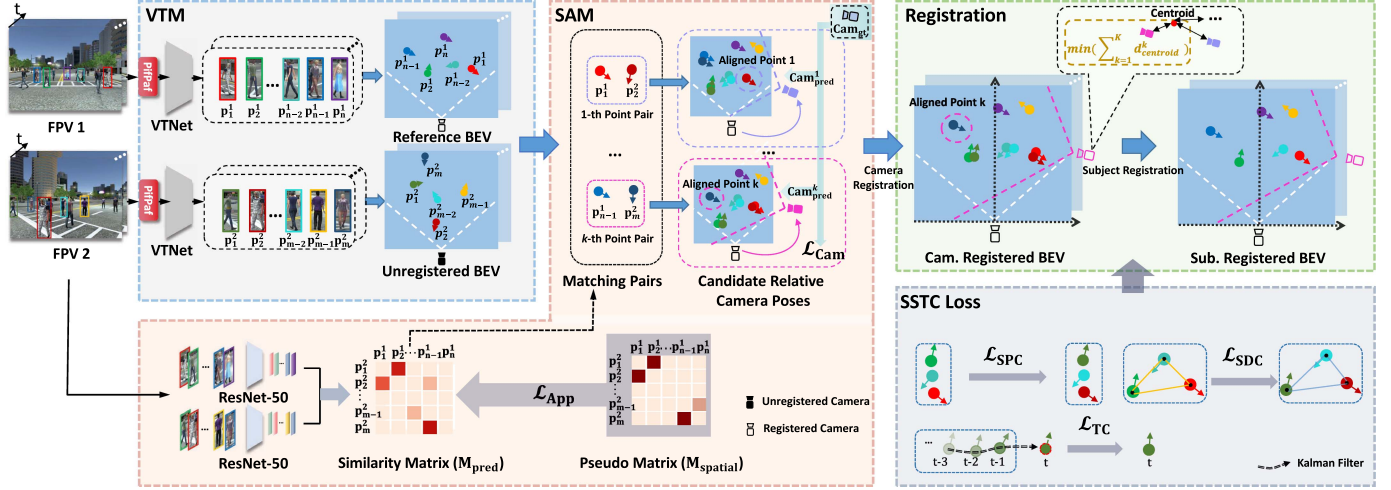


Fig. 2. Framework of the proposed method, which can be divided into three modules (VTM, SAM, and Registration module), and the SSTC learning loss. We use hollow camera icons to represent registered cameras and filled camera icons to represent unregistered cameras.

tracking in the BEV videos captured by the camera mounted on a drone. In [48], [49], such idea is employed to locate the first-person-view camera in BEV aerial images in a large field, which is used for GEO-localization. Later, some related works [5], [50], [51], [52], [53], [54] focus on the localization of humans, which aims to associate and track the multiple humans by the spatial reasoning based on the pre-acquired detection. Another series of works [55], [56], [57] use graph matching-based methods to locate camera wearers by combining information from FPVs and BEV. The main difference between the previous works and this work is that they require a BEV image (e.g., captured by a UAV) as input, which is not practical in many real applications.

III. PROPOSED METHOD

A. Problem Formulation and Method Overview

We first give an overview of the proposed method. As shown in Fig. 2, given multiple videos simultaneously captured from different views for a multi-human scene, we apply a view-transform subject detection module (VTM) to get the position and the face orientation estimation of each person in the BEV (Section III-B1). We then apply a geometric principle based spatial alignment module (SAM) to estimate the relative camera pose in the BEV (Section III-B2). In SAM, with the subject registration results, we further propose a backward training strategy to learn subject similarity for human association in SAM using the spatial alignment as supervision. To train the network, we propose three self-supervised losses considering both the spatial consistency and temporal continuity of the subjects, which are presented in detail in Section III-C. After training, the proposed model can provide the camera and subject registration (candidate) results in the unified BEV. In the inference stage, we also design the camera-subject collaborative registration refinement and temporal tracking strategy. Specifically, as presented in Section III-E, we first use a simple candidate selection strategy based on the centroid distance, to choose the

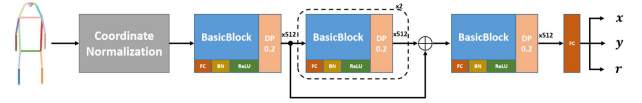


Fig. 3. The structure of VTNet. Here $\times 512$ means the 512 feature channels, FC means the fully connected layer, BN means batch normalization layer, ReLU is an activation function, DP0.2 is the dropout layer with ratio 0.2.

final camera pose estimation result (camera registration) from the candidates obtained by our model, as discussed above. Next, for the subject registration, we take both spatial and appearance information to aggregate the same subject (from different views) in the BEV for subject registration. Finally, in Section III-F, to achieve the BEV tracking, we propose a subject guided camera tracking strategy to track the camera, based on which we design a multi-target tracking scheme for subject tracking.

B. Multi-View Transformation and Alignment in BEV

1) *View-Transform Module (VTM)*: As shown in Fig. 2, given the input of multiple videos captured in a multi-human scene, we first aim to get the subject position and face orientation of each person in the BEV. For this purpose, we first apply an existing human pose detector called PifPaf [58] to predict the 2D skeleton joints of each person from the original RGB image, which will be input into the view-transform module (VTM). The main component of VTM, i.e., VTNet (view-transform network) uses a lightweight FC-based structure with three heads, whose structure is shown in Fig. 3.

At the end of VTNet, we use a human 3D localization head composed of simple MLP layers to predict the position and face orientation of each person. The process of the network can be represented as

$$\mathbf{p}_i^v \triangleq (x_i^v, y_i^v, r_i^v) = \text{VTNet}(\mathbf{k}_i^v), \quad (1)$$

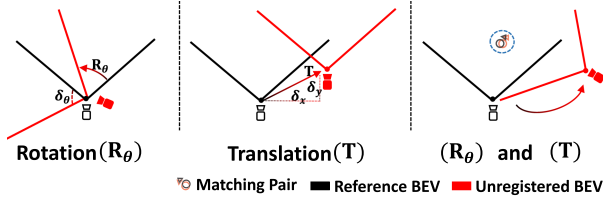


Fig. 4. An illustration of the rotation and translation transformation for two BEVs with a matching pair.

where v denotes the v -th view and i denotes the i -th person in v -th view, $\mathbf{k}_i^v$ is the 2D skeleton joints belonging to the person i in view v . $\mathbf{p}_i^v$ is the output prediction of VTNet, which is composed of (x_i^v, y_i^v) representing the transformed subject position in the BEV and r_i^v representing the face orientation, as shown in VTM of Fig. 2.

2) *Spatial Alignment Module (SAM). Multi-view spatial alignment:* We then show the relative camera pose estimation (in the BEV) via the subject localization alignment. For convenience, we first present the case of two views. Our basic idea is that the human position and face orientation are unique in the real-world 3D coordinate system, which can be used for aligning the cameras that generate multiple 2D images. In the BEV maps with human position and face orientation generated from different first-person view (FPV) images, we can obtain the camera pose in the BEV by aligning the corresponding human position and facing orientation (as aligned points).

For this purpose, the first step is to find the same subject from different views. We first identify the subjects in the input images through the human appearance features, and the corresponding subjects across different views are then matched. Specifically, as shown in SAM of Fig. 2, we use a ResNet-50 network to extract the feature of every person and apply Euclidean distance and sigmoid function to create a similarity matrix ($\mathbf{M}_{\text{pred}}$), which indicates the subject similarities among the subjects from two views. This way, as shown in Fig. 2, the subjects in two BEVs with the same color denote the same subject, which is formed as the point pair. Then we sort the similarities of each subject pair and select the top- K pairs as the *matching pairs*.

After that, we apply the geometric transformation to align two BEVs (containing all subjects and cameras on them), which are denoted as a reference BEV map and an unregistered one. Specifically, for a pair of matching points, we apply a geometric transformation, as shown in Fig. 4, to rotate and move the camera position and orientation in the unregistered BEV to that in the reference BEV.

We calculate the relative pose (in the BEV) between two points in a matching pair as discussed above, which are denoted as a reference one $\mathbf{p}_{\text{ref}} = (x_{\text{ref}}, y_{\text{ref}}, r_{\text{ref}})$ and an unregistered one $\mathbf{p}_{\text{unr}} = (x_{\text{unr}}, y_{\text{unr}}, r_{\text{unr}})$. Specifically, the relative pose transformation between them can be formulated as

$$\begin{pmatrix} x_{\text{ref}} \\ y_{\text{ref}} \\ 1 \end{pmatrix} = \mathbf{T}\mathbf{R}_{\theta} \begin{pmatrix} x_{\text{unr}} \\ y_{\text{unr}} \\ 1 \end{pmatrix},$$

$$r_{\text{ref}} = \delta_{\theta} + r_{\text{unr}}, \quad (2)$$

$$\text{where } \mathbf{R}_{\theta} = \begin{pmatrix} \cos \delta_{\theta} & -\sin \delta_{\theta} & 0 \\ \sin \delta_{\theta} & \cos \delta_{\theta} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \mathbf{T} = \begin{pmatrix} 1 & 0 & \delta_x \\ 0 & 1 & \delta_y \\ 0 & 0 & 1 \end{pmatrix}.$$

We denote $\mathbf{R}_{\theta}$ as the rotation matrix with a rotation angle δ_{θ} , and $\mathbf{T}$ is the translation matrix, in which (δ_x, δ_y) is the translation vector.

The corresponding matching pair can be aligned after applying this transformation. This means the transformation matrix is just the relative camera pose between the two cameras in the BEV. Note that, this relative camera pose only contains three degrees of freedom, i.e., the translation and rotation projected into the BEV plane. This way, we can obtain the relative camera pose $(\delta_x, \delta_y, \delta_{\theta})$ by solving the above (2) and get the *analytical solution* as

$$\begin{cases} \delta_x = x_{\text{ref}} - x_{\text{unr}} \cos \delta_{\theta} + y_{\text{unr}} \sin \delta_{\theta} \\ \delta_y = y_{\text{ref}} - x_{\text{unr}} \sin \delta_{\theta} - y_{\text{unr}} \cos \delta_{\theta} \\ \delta_{\theta} = r_{\text{ref}} - r_{\text{unr}} \end{cases} \quad (3)$$

Weak supervision for SAM: As discussed above, we use K point pairs to estimate the relative pose, each of which can generate a relative pose estimation result as shown in SAM in Fig. 2. Then we can apply the camera pose estimation loss function for training our model as

$$\mathcal{L}_{\text{Cam}}^{\text{full}} = \sum_{k=1}^K \left(\|(\delta_x^k, \delta_y^k) - (\delta_x^{\text{gt}}, \delta_y^{\text{gt}})\| + \|\delta_{\theta}^k - \delta_{\theta}^{\text{gt}}\| \right), \quad (4)$$

where $\text{Cam}_{\text{pred}}^k = (\delta_x^k, \delta_y^k, \delta_{\theta}^k)$ is the k -th candidate relative camera pose estimation generated from the k -th point pair, and $\text{Cam}_{\text{gt}} = (\delta_x^{\text{gt}}, \delta_y^{\text{gt}}, \delta_{\theta}^{\text{gt}})$ is the ground-truth relative camera pose. Note that, we apply the supervision *only on the camera* position (δ_x, δ_y) and view direction (δ_{θ}) , which involves *no supervision on the subjects*.

Moreover, we know that the camera view direction is very hard to measure and annotate in real-world applications. In the experiments, we show that our method *is not very sensitive* to the supervision of δ_{θ} . This way, the supervision for SAM in (4) can be further simplified by removing δ_{θ} as

$$\mathcal{L}_{\text{Cam}} = \sum_{k=1}^K \|(\delta_x^k, \delta_y^k) - (\delta_x^{\text{gt}}, \delta_y^{\text{gt}})\|. \quad (5)$$

C. Self-Supervised Spatial-Temporal Consistency (SSTC) Learning Loss

Based on the above VTM and SAM, we then aim to design effective losses to train the whole model. The straightforward supervision is the ground-truth localization of the subjects in BEV. However, this supervision is hard to obtain in practice. We propose to train the subject registration and tracking in a self-supervised manner.

Our basic idea is that, for a unified BEV generated from multiple views, the same person in different views should be located in the same position. Specifically, we consider this prior constraint from two aspects. On the one hand, we consider the consistency from the spatial domain among the different FPVs.

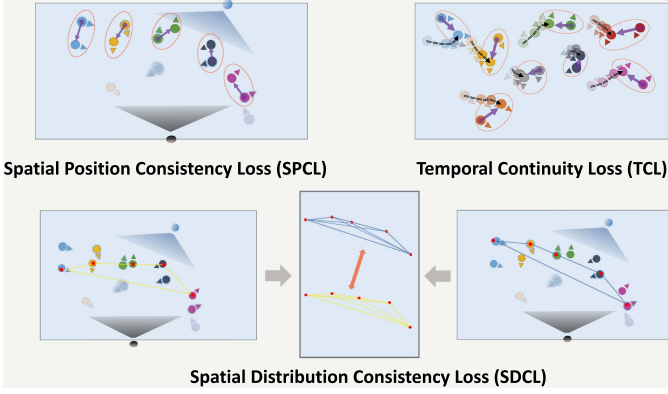


Fig. 5. Illustration of the proposed SSTC loss, including the cross-view spatial losses (SPCL, SDCL), and over-time temporal loss (TCL).

Inspired by the fact that no matter which FPV a person appears in, her/his position under the world coordinate system is unique. So we consider the consistency among multiple views to construct the *Cross-view Spatial Loss*, including SPCL and SDCL, as shown in the SSTC loss in Fig. 2. This involves the position consistency of each subject in different views within a unified BEV coordinate system, as well as the spatial distribution consistency of all the subjects among different views. On the other hand, we consider the continuity in the temporal domain. Specifically, the prediction of a subject in the current time should be adjacent to the prediction of the historical trajectory from previous frames, i.e., the predicted and detected positions should be continuous, which forms the *Over-Time Temporal Loss*, i.e., TCL as shown in Fig. 2. We then describe the above self-supervised losses in detail.

① *Spatial position consistency loss (SPCL)*: First, as shown in Fig. 5, in the unified registered BEV, we use the appearance similarity to identify the same person from different side views. Then, we measure the subject position consistency loss between views u and v as

$$\mathcal{L}_{\text{SPC}}^{u,v} = \sum_{j=1}^N \|(x_j^u, y_j^u) - (x_j^v, y_j^v)\|, \quad (6)$$

where $\|\cdot\|$ denotes the 2-norm for the vector. j represents the j -th pair of identical targets in all N pairs, and (x_j^u, y_j^u) , (x_j^v, y_j^v) denote their coordinates in the BEV, respectively. When the identical targets across views are closer in position under a unified BEV coordinate system, their subject position consistency will be higher, making this loss lower.

② *Spatial distribution consistency loss (SDCL)*: Besides the position consistency loss for each target, we also consider a global spatial distribution consistency loss as below. Specifically, we also take two views u and v as an example. For view u with N subjects, we calculate the distance between each subject pair in this view to obtain a distance matrix with dimensions $N \times N$, which is normalized to obtain the normalized distance matrix $\mathbf{M}_u \in \mathbb{R}^{N \times N}$. For the other view, we perform the same operation on the same subjects to obtain another normalized distance matrix $\mathbf{M}_v$. The subject distributions between view u

and view v should be consistent, which is formulated as the following loss

$$\mathcal{L}_{\text{SDC}}^{u,v} = \text{MSE}(\mathbf{M}_u - \mathbf{M}_v), \quad (7)$$

where $\text{MSE}(\cdot)$ denotes the mean square error to measure the matrix difference.

Clarification: The SPC loss is designed to constrain the spatial relationship at the individual level, while the SDC loss here is to consider the global spatial relationship of the subjects. These two self-supervised cross-view spatial losses can simultaneously help train the network for learning cross-view spatial relationships and predict the subject positions more accurately in the unified BEV.

③ *Temporal continuity loss (TCL)*: Besides the cross-view spatial consistency for each time, we also consider the temporal continuity loss along the sequence. Specifically, for each side view, we first track all subjects in the previous l frames, e.g., 10 frames, and generate multiple tracklets, where we use the similarity of appearance features. Then, in the BEV view, we use the motion tendency, e.g., a linear motion model, to predict the position of these subjects in the current frame based on the previous tracklets. Since it is intuitive that, in BEV, the transformed subjects in the current frame should be continuous with those predicted by the tracklets in previous frames. This way, we propose the continuity loss between the positions of the transformed subjects and those predicted by the tracklets, satisfying the following loss

$$\mathcal{L}_{\text{TC}}^{v,t} = \sum_{k=1}^M \|(x_k^v, y_k^v) - (\tilde{x}_k^v, \tilde{y}_k^v)\|, \quad (8)$$

where v is v -th view, t denotes the frame t , and M is the number of matched subjects and tracklets, k is the k -th match. In view v at time t , (x_k^v, y_k^v) is coordinate of the k -th transformed subject obtained by VTNet in (1), and $(\tilde{x}_k^v, \tilde{y}_k^v)$ is the predicted coordinate of the k -th subject based on the previous tracklet. We can match the current subjects with previous tracklets using the classical multi-object tracking (MOT) approach without annotation. We clarify that the TCL is calculated separately for each view. The losses for all views and frames are superimposed to obtain the final loss $\mathcal{L}_{\text{TC}}$. The temporal loss can help the BEVTrans have better continuity on the tracking task.

④ *Subject alignment feedback supervision*: Besides the above subject alignment supervisions (to train the VTNet), we further consider to use them for supervising the appearance-based subject association network with a back-propagation strategy. As shown in the bottom of Fig. 2, we propose to train the appearance feature extraction network, i.e., ResNet-50, for similarity measurement in a self-supervised manner, to make full use of the spatial alignment.

On one hand, similar to SPC loss, we create a person spatial distance matrix $\mathbf{M}_{\text{dis}}$ and an angle difference matrix $\mathbf{M}_{\text{ang}}$ in the unified registered BEV, which measure the distance and angle differences of all subjects from different views. We inversely normalize each row of the spatial distance matrix $\mathbf{M}_{\text{dis}}$ and angle difference matrix $\mathbf{M}_{\text{ang}}$ discussed above, to get the spatial-aware similarity matrixes as $\mathbf{M}_{\text{spatial}} = \lambda \overline{\mathbf{M}_{\text{dis}}} + (1 - \lambda) \overline{\mathbf{M}_{\text{ang}}}$, where

λ is the hyper-parameter, $\overline{M}_{dis}$ and the $\overline{M}_{ang}$ are the normalized similarity matrix obtained from M_{dis} and M_{ang} , respectively. On the other hand, we use the obtained appearance similarity matrix (M_{pred}) indicating the subject similarities among the subjects from different FPVs, as discussed above.

This way, we apply a feedback-supervised loss as below, to make use of the spatial alignment for training the appearance feature extraction network in our model

$$\mathcal{L}_{App} = \|M_{pred} - M_{spatial}\|. \quad (9)$$

D. The Training Framework

During training, we first pre-train VTNet in VTM using the subject locations and directions with our synthetic dataset, optimized with MSE loss. This step equips the model with basic view transformation capabilities under a single-view setting. In the main training stage, no subject annotations are used and all subject-related losses are self-supervised. This stage involves training all components of the network using only the camera location annotations in (5) combined with the proposed SSTC losses in Section III-C, i.e., the total loss function

$$\mathcal{L} = \mathcal{L}_{Cam} + \mathcal{L}_{SPC} + \mathcal{L}_{SDC} + \mathcal{L}_{TC} + \mathcal{L}_{App}. \quad (10)$$

Note that, in the main training process, we only use the camera localization of the FPVs as supervision, which is significantly economical for training data annotation for this problem.

Training losses for downstream tasks: BEVTrack contains two main tasks, i.e., 1) registration and 2) tracking in the BEV of both FPV cameras and subjects. We discuss the relationship between the losses and the downstream tasks here.

First, for the registration task, which includes both camera registration and subject registration, we employ multiple loss functions. For camera registration, we directly apply the camera location loss $\mathcal{L}_{Cam}$ for supervision. Note that, this is the only supervised loss that requires annotations in our method. For subject registration, we have applied both the Spatial Position Consistency Loss (SPCL), i.e., $\mathcal{L}_{SPC}$ and Spatial Distribution Consistency Loss (SDCL), i.e., $\mathcal{L}_{SDC}$, that work together to ensure accurate multi-view subject registration while improving camera registration reliability. Moreover, as shown in Fig. 2, we also apply the self-supervised appearance similarity learning loss $\mathcal{L}_{App}$ in our method. On one hand, the appearance based subject matching is the prerequisite for the camera registration. On the other hand, the subject registration results can in turn supervise the appearance learning.

Second, for the tracking task, the Temporal Consistency Loss (TCL) i.e., $\mathcal{L}_{TC}$, captures subtle position changes of targets in motion, ensuring continuity of target localization during tracking, thereby significantly improving overall tracking performance.

In summary, these loss functions for each downstream task are integrated into a final loss function (10) for network training. They can create a robust multi-task learning framework where each loss contributes to specific downstream tasks while also providing mutual enhancement through multi-task complementarity.

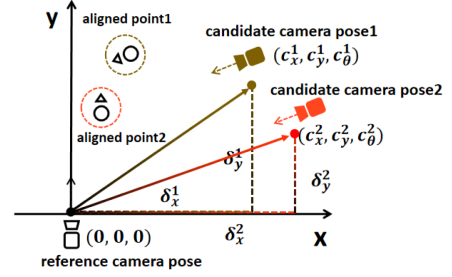


Fig. 6. Candidate camera poses in the coordinate system.

Advantage discussion: Here we discuss about two main advantages of the proposed framework. 1) *Simple network structure:* In our model, the main network parameters exist in the MLP in VTM and the feature extraction network in SAM. Such a simple and lightweight network is easy to deploy in practice. 2) *Economical supervision annotation.* More importantly, although the desired output of our method is ample, including the spatial registration and temporal tracking of both subjects and cameras, the required supervision of our model is economical. As presented before, the only annotation for supervision is the camera localization in (5). Other losses do not require the annotation such as the cross-view subject matching and the subject BEV localization. This greatly reduces the training costs since these annotations are not easily accessible.

We then elaborate on the registration and tracking in BEV for camera and subject during the inference stage, in the following.

E. Camera and Subject Registration in BEV

As discussed above, on one hand, *accurate FPV camera localization and view direction estimation are crucial for the subject registration in BEV.* So we first consider the camera registration in BEV.

Camera registration in BEV: Based on the relative camera pose $(\delta_x^k, \delta_y^k, \delta_\theta^k)$ obtained from our model (as discussed in Section III-B2), we have got K candidates of relative camera pose estimation between the reference and unregistered BEVs. Here we denote the camera pose on the reference BEV as $(0, 0, 0)$. We then get a camera pose of the unregistered BEV in the coordinate system of the reference BEV as

$$c^k = (c_x^k, c_y^k, c_\theta^k) = (0, 0, 0) + (\delta_x^k, \delta_y^k, \delta_\theta^k) = (\delta_x^k, \delta_y^k, \delta_\theta^k), \quad (11)$$

which denotes the k -th candidate camera pose from $(\delta_x^k, \delta_y^k, \delta_\theta^k)$, as shown in Fig. 6.

The next step is to find the selected camera pose from the K candidates, i.e., achieving the camera registration task. We calculate $\bar{c}_x = \frac{\sum_{k=1}^K c_x^k}{K}$, $\bar{c}_y = \frac{\sum_{k=1}^K c_y^k}{K}$, where (c_x^k, c_y^k) is the candidate position of the unregistered camera, $(\bar{c}_x, \bar{c}_y)$ is the centroid point. We then compute the distance of each candidate position to the centroid point as

$$d_{centroid}^k = \|(c_x^k, c_y^k) - (\bar{c}_x, \bar{c}_y)\|, \quad (12)$$

The candidate with the minimum distance will be selected, which is used to register the unregistered BEV into the reference

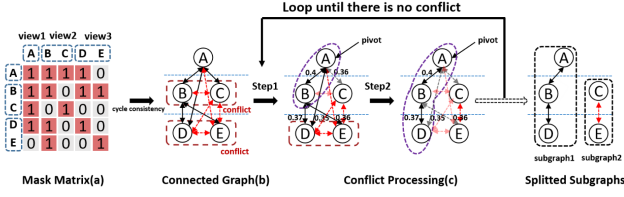


Fig. 7. Solving the cycle consistency and uniqueness.

BEV, then we can get a unified *camera registered BEV* as shown in the left part of Registration module in Fig. 2.

Subject registration in BEV: With the camera registration result, we can register the camera position and its view direction, together with the subject localization and face orientation of the unregistered BEV, into the reference BEV. Note that, for multiple views, we select one as the reference BEV and others as the unregistered BEVs, all of which can be registered into the reference BEV, respectively. The next step is aggregating the same person from different views in the registered BEV, which can be achieved by two steps, i.e., subject matching and fusion.

① **Subject matching stage:** We use the spatial similarity matrix M_{dis} , M_{ang} and the appearance similarity matrix M_{pred} provided in Section III-B2, for subject matching. We employ three thresholds on the above matrices, respectively, as filters to select potential matching subject pairs, whereby only pairs that fall within the distance and angle thresholds and surpass the similarity threshold will be identified as the same subject. Besides, we further consider two constraints for accurate matching. The first one is cycle consistency [59], which means the connection of the same subject from all views should form a loop. The second one is uniqueness, which means one subject should not be connected to more than one subject in another view. They are implemented as below.

① We use a classical data structure, i.e., union-find, to aggregate the transitive relations, which makes all the subjects with direct and indirect connections in a union of union-find to be clustered as a sub-graph, as shown in Fig. 7(b), which solves the problem of cycle consistency for all the subject connected as a loop. ② We define the problem as a hierarchical maximum spanning subgraph problem, the layer-by-layer (view-by-view) spanning constraints that a subject is connected at most one node in each view to avoid the uniqueness conflict, as shown in Fig. 7(c). To solve this problem, we propose an algorithm referenced from the Prim algorithm [60]. We provide more details and the algorithm flow of the above strategy in the *supplementary material*.

② **Subject fusion stage:** For the subjects from multiple views to be regarded as the same person using the above subject matching method, we then estimate the final registration result of a subject using the same strategy in (12). The position and orientation of the subject with the minimum centroid distance will be retained, and other same subject will be removed. Especially, if there are only two views, we use the mean position and orientation between two subjects as the fused result. Finally, we can get the final *subject registered BEV* with the camera and subject registration from multiple views.

F. Camera and Subject Collaborative Tracking

In the above section, we have presented the camera and subject registration in BEV. In Section III-C, we make use of the spatial-temporal information used in the training phase. Similarly, we also take into account spatial-temporal guidance during the inference stage for BEV tracking in this section.

Camera tracking: Similar to Section III-E, we also first consider the camera tracking. A straightforward approach is to estimate the camera's localization/direction through BEV alignment in Section III-E frame by frame. However, such method relies on the accuracy of aligned subjects. The errors from the false alignment or imprecise detection may accumulate and affect the accuracy of camera localization/direction estimation in each frame. For a better and smoother result, we handle the camera location and direction tracking in the video by considering the spatial-temporal information.

① **Camera location tracking:** On one hand, we can utilize the camera locations from the previous τ moments to predict the camera position at the current time T (temporal cue). On the other hand, we can directly estimate the camera location at time T with the method in Section III-C (spatial cue). We propose a fusion strategy for the final camera localization as below.

1) **Temporal Cue for Camera Location Tracking:** We introduce a prediction model that is based on both temporal continuity and multi-view subject position consistency to predict the current frame's camera position (c_x^T, c_y^T) as

$$\begin{cases} c_x^T = \sum_{t=T-\tau}^T \alpha^t (\Delta t (c_x^t - c_x^{t-1}) + c_x^t) \\ c_y^T = \sum_{t=T-\tau}^T \alpha^t (\Delta t (c_y^t - c_y^{t-1}) + c_y^t) \end{cases}, \quad (13)$$

where (c_x^t, c_y^t) denotes the predicted camera position at history moment t based on the temporal prediction model. Specifically, $\Delta t = T - t$ denotes the time gap for each moment t , we use $(c_x^t - c_x^{t-1})$ to predict the velocity rate factor (at x axis) at time t . The current localization is predicted by all the previous τ moments, i.e., $t = T - \tau, \dots, T$. α^t denotes the confidence factor at time t as $\alpha^t = \frac{f^t}{\sum_{t=T-\tau}^T f^t}$, in which we define $f^t = \frac{1}{\beta C_{SPC}^{t-1} + (1-\beta) C_{SPC}^t}$. Our basic idea is that the higher spatial position inconsistency (calculated by SPC score defined in (6)) leads to a lower f^t , therefore the lower confidence factor α^t , which will help the prediction model better balance the rates for each time interval. Here β is a hyperparameter used to balance the weights of SPC scores between adjacent moments.

2) **Spatial-Temporal Fusion for Camera Tracking:** After obtaining the predicted camera position, we adopt the following fusion strategy by considering both the camera location (c_x^G, c_y^G) obtained through the BEV transform and alignment in the current frame, and the temporally predicted location (c_x^T, c_y^T) . The final camera location at the current moment is determined as

$$(c_x, c_y) = \gamma (c_x^G, c_y^G) + (1 - \gamma) (c_x^T, c_y^T), \quad (14)$$

where γ is a weighting factor used to balance the importance between two results. To avoid the false correction, if the distance between (c_x^G, c_y^G) and (c_x^T, c_y^T) exceeds a threshold value η , it is considered that the result of the spatial cue is inaccurate.

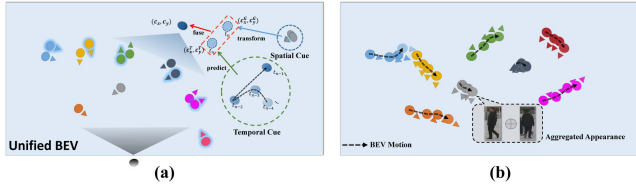


Fig. 8. An illustration of the multi-view temporal detection for camera tracking (a) and the proposed BEV motion and multi-view aggregated appearance features (b).

This way, we directly use (c_x^T, c_y^T) as the final camera location (c_x, c_y) . The overall flow discussed here can be seen in Fig. 8(a).

② *Camera view direction tracking*: For the camera view direction, we also consider both the spatial and temporal information for tracking. Specifically, we record the view directions of the camera from previous τ moments and the direction computed from the current spatial alignment. Then, we select the view direction with the highest SPC score as the final camera direction, denoted as c_θ . At present, we obtain the camera location and view direction (c_x, c_y, c_θ) tracking results in the unified registered BEV along the whole video.

Subject tracking: Based on the camera tracking strategy as discussed above, we obtain more accurate FPV camera location/direction. This way, we can construct a more smooth and stable BEV, in which we have accurate and reliable subject detection results. Also, as discussed above, the camera tracking is with the help of the spatial distribution of the subjects in the camera's view. Thus we clarify that *the camera and subject tracking are complementary and collaborative*, which can help each other for better performance. Then, based on this reliable BEV, we further consider new features suitable for tracking subjects in the BEV to obtain the final result of BEV tracking, as below.

For multi-subject tracking, the most commonly used features are appearance and motion [4], [5]. Instead of using the traditional motion and appearance features obtained from bounding boxes in the single FPV view, we propose the novel BEV motion feature obtained from the subject position in the fused BEV, and the aggregated appearance features merged from multiple views, as *the new features for multi-view multi-subject BEV tracking*.

① *The new motion feature in BEV*: Rather than using the IOU and distance of past detection boxes in the 2D image for motion feature extraction, we construct motion features based on the position and orientation of subjects in the generated BEV, i.e., the *BEV motion feature*. As illustrated in Fig. 8(b), the proposed motion features based on the spatial information of objects in the BEV, have more clear trajectories, which are more continuous and steady without the error due to 2D projection transformation.

② *The new aggregated appearance feature*: We obtain the fused BEV where the subjects are the fusion of objects from multiple FPVs. This way, each subject in the BEV contains appearance feature information from multiple views, as shown in Fig. 8(b). We modify the conventional approach of using FPV detection boxes. The appearance feature is computed as the average of features from all FPVs, i.e., the *aggregated appearance feature*.

In our method, we combine the above new motion and appearance features from multi-view for each subject, effectively improving the multi-view BEV tracking.

G. Implementation Details

Parameter settings: In the SAM, we use a ResNet-50 based pre-trained model in [61] for appearance feature extraction. In Section III-B2, we set the number of candidate K as 3, and set the hyper-parameters of the pseudo matrix as $\lambda = 0.5$. In Section III-E, we set the similarity matrix threshold as 0.25, the distance threshold as 2.0 m and the angle threshold as 15° . In Section III-F, we set the hyperparameter $\beta = 0.4$, $\gamma = 0.3$, and the threshold value $\eta = 0.5$, through grid hyperparameter search.

Training and inference settings: We use Pytorch as our main framework and the work runs on the server with RTX 3090 GPU. During training, we use the pairs of FPVs as input to train our framework. In the inference stage, the number of FPVs is not limited, in which one FPV will be selected as the reference view and others can be registered in that reference view. For the tracking framework, we adopt the widely used StrongSORT [62], which is a strong and fair baseline for MOT. Note that, with the newly proposed appearance and motion features, the tracking is achieved with the unchanged time complexity compared to StrongSORT.

IV. BEVTRACK BENCHMARK

First, we use a public multi-view human tracking dataset CVMHAT [5] with some necessary modifications for our task. In CVMHAT, besides the side views provided by the wearing cameras, they also capture the real BEV using a drone. It also provides the annotations of the bounding boxes for each subject and assigns them a unified ID number for all the FPV and BEV videos. Therefore, the videos and annotations can be applied to BEVTrack. Specifically, CVMHAT includes five different scenes. For each scene, the number of individuals ranges from 7 to 12. Since CVMHAT only contains the pair-wise videos in their data splitting, we reconstruct this dataset as CVMHAT-BEVT. CVMHAT-BEVT contains a number of side views ranging from 2 to 4 in each scene. In total, CVMHAT-BEVT includes 21 synchronous multi-view videos with a duration ranging from 200 to 1,500 frames.

Since CVMHAT-BEVT is not very large, we use it as an evaluation dataset in this work. However, besides CVMHAT, we can not find a directly available dataset for our task. For example, existing datasets for multi-view multi-human tracking, e.g., MvMHAT [3], DukeMTMCT [31], do not provide the available annotations of the multi-view camera poses, human positions in BEV. These annotations are difficult to achieve in the real world. Although a few synthetic datasets [7], [19] include these annotations, it is not a video dataset. Each frame is randomly generated, making it unsuitable for the tracking task discussed in this work. Therefore, we consider using a modeling engine to create a large-scale synthetic dataset for training and evaluations of BEVTrack.

Specifically, we use Unity 3D engine and PersonX model library [63] to simulate various real-world scenes. To ensure diversity and randomness in the dataset, we have constructed five different scenes, each containing a different number of walking persons ranging from 15 to 20. The moving direction of each subject is free, ensuring that the walking trajectories of each person are various. We also include several moving subjects carrying cameras for FPV video capturing. The synthetic scene covers $25\text{ m} \times 25\text{ m}$ scaled to real-world proportions. Each scene consists of a video with 1,000 frames. The proposed synthetic dataset called *BEVT-S* contains the position (in meters) and body orientation of each subject in the BEV, as well as the camera poses. Besides, we also provide the bounding box with the unified ID number of each subject in all views. For each scene, we generated 15 pairs of FPV videos, resulting in a total of 75 video pairs. These video pairs are then divided into non-overlapping sets, with 50 used for training and 25 video pairs used for testing. For the synthetic dataset, we consider the simplest setting of two views, which can be easily extended to more views. This large dataset can be used for both training and testing. We show the examples of the dataset in the *supplementary material*. We also consider the cross-domain setting from the synthetic dataset to the real dataset, which is presented in detail in the experiments.

V. EXPERIMENTS

A. Evaluation Metrics

To evaluate our proposed method quantitatively, we define the following metrics.

- *Metric-I*: We first evaluate the accuracy of the detection of camera results, which includes the average localization error (Cam.Loc.) in meter and average view direction error (Cam.Dir.) in degree of the camera in BEV.
- *Metric-II*: We evaluate the accuracy of the detection of subject results, which includes the average position error (Sub.Pos.) in meter and average body orientation error (Sub.Ori.) in degree of the subjects in the scene.
- *Metric-III*: We also evaluate the tracking-related metrics in BEV, including the traditional metrics MOTA [68] and IDF_1 [31], as well as the recent comprehensive metric HOTA [69]. Note that, we evaluate the tracking results in the BEV. Due to the absence of a standard ground-truth BEV that is aligned to the generated unified BVE, we cannot use the traditional approach of constructing a similarity matrix using the IOU of detection boxes. Instead, we construct the similarity matrix with the normalized distance and then use the Hungarian algorithm for matching. We evaluate the performance based on the matching results. The overall evaluation process is the same as the default setting in TrackEval [70], except for the difference in constructing the similarity matrix.

B. Comparative Results

Comparison methods: Since there are **no** existing methods specifically designed to address the proposed new BEVTrack

task, we try to conduct as many related approaches as possible for comparison.

- *BEVSee*: We compare the method [19] for multi-view BEV detection by adding the tracker [62] in our method.
- *Monoloco++*: We compare the single-view BEV detection method Monoloco++ [64] by adding the BEV alignment and the tracker. Monoloco++ is a network trained on KITTI and nuScenes datasets, which is used to predict the 3D-localization and face orientation of each person in the view. We concatenate it with our proposed geometric transformation and subject fusion methods for evaluation. Besides, for the sub-task of camera registration, we also include the following three methods.
- PatchNet [18], CVNet [65] and R2Former [66] are local descriptor (key point) matching based methods, which are combined with the classical camera pose estimation method with the matched key points for relative camera estimation. The input of these methods is a pair of FPV images and then we can get some key point matching pairs. After that, we use the classical camera pose estimation method with the matched key points to generate the essential matrix and convert it to the relative camera location and (yaw-axis) direction.

For the sub-task of subject registration, we include two methods for multi-view detection.

- *MVDet* and *MVDetr* [7], [22] are used for multi-view object detection with the camera calibrations. These two methods need camera calibrations to generate the results of subject registration. So we calculate the camera calibrations by using the 3D localization of feet (with height = 0) and the 2D position of the bottom of the bounding box of each person predicted in our methods. With the calibration, these two methods generate multi-view human detection predictions (without human identifications) in the BEV. Then we evaluate the results by using the Hungarian matching algorithm to match the identification of all the predicted points with the ground-truth ones through the minimum spatial distance.

We also compare with several tracking methods by adding VTM in our method to get the targets' BEV positions as below.

- *ReST*, *MvMHAT*, *CrossMOT*, and *CRTracker* [28], [29], [30], [67] are recent multi-view tracking methods. We perform multi-view tracking directly in multi-side views, and assigns the tracked IDs directly to the corresponding targets in the BEV. All these methods are adapted to work with our VTM module to generate BEV positions for fair comparison.
- *VIEW_COMBINE*: Apply the tracking in each view followed by the subject association among different views, and then aggregate the same subject IDs.
- *VIEW_MAP*: Apply the tracking in each view, and map to BEV, respectively. The tracking performance is calculated by averaging the results in all views.

Camera & subject registration results: We first evaluate the camera and subject registration results on the BEVT-S dataset as shown in the left four columns in Table I.

TABLE I

COMPARISON RESULTS ON BEV-T-S, INCLUDING THE ACCURACY OF CAMERA REGISTRATION USING THE METRICS OF CAM.LOC. AND CAM.DIR. (METRIC I), THE ACCURACY OF SUBJECT REGISTRATION USING SUB.POS. AND SUB.ORI. (METRIC II), AND THE ACCURACY OF SUBJECT TRACKING USING HOTA, MOTA, AND IDF_1 (METRIC III).

Methods	Cam.Loc. ↓	Cam.Dir. ↓	Sub.Pos. ↓	Sub.Ori. ↓	HOTA ↑	MOTA ↑	IDF_1 ↑
BEVSee [19]	1.77 <i>m</i>	8.19°	1.11 <i>m</i>	15.76°	59.36%	70.36%	60.32%
Monoloco++ [64]	3.22 <i>m</i>	13.48°	0.97 <i>m</i>	20.39°	43.34%	69.65%	55.09%
PatchNet [18]	14.28 <i>m</i>	78.17 °	-	-	-	-	-
CVNet [65]	14.53 <i>m</i>	90.15 °	-	-	-	-	-
R2Former [66]	13.82 <i>m</i>	84.36 °	-	-	-	-	-
MVDet [7]	-	-	2.85 <i>m</i>	-	-	-	-
MVDetr [22]	-	-	2.70 <i>m</i>	-	-	-	-
ReST [28]	-	-	-	-	64.23%	68.73%	72.70%
MvMHAT [67]	-	-	-	-	65.36%	69.94%	73.94%
CrossMOT [29]	-	-	-	-	71.73%	67.30%	74.05%
CRMOT [30]	-	-	-	-	73.05%	68.23%	75.53%
VIEW_COMBINE	-	-	-	-	61.30%	64.74%	72.02%
VIEW_MAP	-	-	-	-	66.83%	72.30%	77.23%
Ours	0.77 <i>m</i>	3.04°	0.84 <i>m</i>	10.94°	78.13%	77.03%	82.61%

Regarding the camera registration metrics, we can first see that all the comparison methods provide very poor results. The key point matching based methods PatchNet, CVNet and R2Former almost fail because of the huge view differences. Monoloco++ generates a relatively acceptable result since it's equipped with the proposed geometric transformation method VTM. We can see that our method outperforms other comparative approaches by achieving a detection accuracy with an error of fewer than 0.77 meters, less than 1 m, surpassing the image-based method of our preliminary work [19] by about 1 m. Also, our method exhibits an approximate 5° improvement from 8.19° to 3.04° in view direction estimation.

In terms of the subject registration, we evaluate the results using Metric-II as shown in the middle of Table I. Similar to the camera registration results, our method outperforms all the comparative approaches. Specifically, our method provides better performance than our preliminary work BEVSee [19]. Particularly, in terms of orientation angle error, where we achieve an improvement of about 5°. Even MVDet and MVDetr take the camera calibration as prior, our method achieves much superior results than them. It keeps the average distance error of 0.84 *m* within 1 m and the average orientation error about 10 degrees. This is promising for many real-world applications.

Subject tracking results: For the tracking metrics, our method obtains the highest performance among all comparative methods in MOTA and IDF_1 and the comprehensive HOTA metrics, with large margins, which demonstrates the superiority of our method. Specifically, compared to the most recent methods, our approach achieves significant improvements: 5.1% higher HOTA than CRMOT [30], 6.4% higher than CrossMOT [29], 12.8% higher than MvMHAT [67], and 13.9% higher than ReST [28]. These results demonstrate the effectiveness of our self-supervised spatial-temporal consistency learning approach compared to state-of-the-art multi-view tracking methods. Note that, for the last six methods, we apply the VTM in our method to help them.

C. Ablation Study

We also consider several variations of our method in the following.

- $w \delta_{\theta}^{gt}$: In SAM, adding the supervision of the camera orientation as in (4).
- *w/o SPC/SDC/TC loss*: ss in (6)–(8), respectively, in Section III-C.
- *Max/Random selection*: In the candidate camera selection strategy, we choose the max confidence pair or choose randomly instead of our method in (12) for camera registration in Section III-E.
- *w/o camera tracking*: Removing the spatial-temporal cues for camera tracking in Section III-F, i.e., only considering frame-by-frame camera registration results without tracking cameras.
- *w Kalman filter in camera tracking*: Using classical Kalman filter to replace the temporal camera position prediction in (13).
- *w/o SPC in camera tracking*: Removing the SPC for temporal cues construction in Section III-F by simply setting $\alpha^t = \frac{1}{T}$ using averages in (13).
- *w/o motion feature in subject tracking*: Removing the motion feature for subject tracking in Section III-F, only using aggregated appearance feature for tracking.
- *w/o appearance feature in subject tracking*: Removing the appearance feature for subject tracking in Section III-F, only using motion feature for tracking.
- *w/o appearance feature aggregation in subject tracking*: Removing the aggregated appearance feature for subject tracking in Section III-F, only using the single-view appearance feature.

We first investigate the effectiveness of some components in the training stage. First, as shown in Table II, we can first see that, when adding the additional camera orientation supervision in SAM, i.e., ' $w \delta_{\theta}^{gt}$ ', the performance does not change considerably. This demonstrates that our method *is not heavily dependent on* the camera orientation supervision, which is not

TABLE II
ABLATION STUDY RESULTS ON BEVT-S BY REMOVING/REPLACING SOME KEY COMPONENTS IN OUR METHOD

Methods	Cam.Loc. ↓	Cam.Dir. ↓	Sub.Pos. ↓	Sub.Ori. ↓	HOTA ↑	MOTA ↑	IDF ₁ ↑
w $\delta_{\theta}^{\text{gt}}$	0.79 <i>m</i>	2.91°	0.90 <i>m</i>	10.28°	78.57%	76.99%	83.36%
w/o SPC loss	1.83 <i>m</i>	6.89°	0.99 <i>m</i>	14.14°	75.16%	75.30%	80.37%
w/o SDC loss	1.05 <i>m</i>	4.02°	0.90 <i>m</i>	14.39°	76.84%	76.61%	81.24%
w/o TC loss	1.15 <i>m</i>	4.23°	0.89 <i>m</i>	12.77°	75.72%	74.38%	79.97%
Max selection	1.72 <i>m</i>	8.04°	1.15 <i>m</i>	18.42°	61.10%	71.88%	62.41%
Random selection	1.60 <i>m</i>	7.45°	1.14 <i>m</i>	16.13°	65.02%	73.68%	67.62%
w/o camera tracking	1.36 <i>m</i>	6.46°	1.01 <i>m</i>	13.99°	66.74%	73.39%	65.81%
w Kalman filter in camera tracking	1.40 <i>m</i>	6.60°	1.05 <i>m</i>	14.94°	73.29%	71.57%	78.05%
w/o SPC in camera tracking	1.14 <i>m</i>	4.56°	0.99 <i>m</i>	12.75°	76.92%	75.39%	80.18%
w/o motion feature	0.77 <i>m</i>	3.04°	0.84 <i>m</i>	10.94°	71.29%	68.57 %	78.05 %
w/o appearance feature	0.77 <i>m</i>	3.04°	0.84 <i>m</i>	10.94°	65.90%	71.68 %	68.58%
w/o appearance feature aggregation	0.77 <i>m</i>	3.04°	0.84 <i>m</i>	10.94°	76.96%	75.97%	79.57%
Ours	0.77 <i>m</i>	3.04°	0.84 <i>m</i>	10.94°	78.13%	77.03%	82.61%

easy to obtain in the real world. Then, we can see that removing the spatial-related SPC and SDC losses decreases the accuracy of the camera and subject localization. We find that the SPC loss has a more significant impact on the results than the SDC loss. This is natural since the SPC loss is the direct loss for spatial supervision. But the SDC loss is also useful as a complementary loss. Also, removing the TC loss, which captures temporal relationships, has a more significant impact on tracking results. This ablation study leads to the lowest MOTA and IDF₁ scores among them, which indicates that the TC loss containing the temporal information is significant and effective for the tracking task.

We then consider about the ablation study in the registration stage. Specifically, for camera pose selection in the camera registration module, we can see that no matter whether using the strategy of the max confidence ('Max') or the random one ('Random'), which, not considering the spatial-aware selecting strategy, both provide a relatively poor performance than our centroid strategy.

Next, our ablation study on camera tracking demonstrates that when camera tracking is removed, there is a significant degradation in both camera-subject registration and subject tracking performance. More specifically, the results reveal that the Kalman filter, which disregards scene spatial information, suffers from cumulative errors, leading to substantially inferior camera-subject registration and tracking performance compared to our method. Also, the proposed SPC effectively leverages spatial information of scene subjects, enhancing camera-subject registration accuracy. These findings validate that the spatial-temporal cues in camera tracking play a crucial role in improving overall BEVTrack performance. Finally, our ablation studies on subject tracking demonstrate that both the proposed new motion and aggregated appearance features significantly enhance tracking performance. Notably, integrating the new motion feature shows substantial improvements across all tracking metrics compared to using appearance features alone. Additionally, the aggregated appearance feature effectively improves the robustness of subject representations, improving tracking results compared to the single-view appearance. It is worth mentioning

TABLE III
ANALYSIS ON THE EFFECTIVENESS OF SUBJECT ALIGNMENT FEEDBACK SUPERVISION STRATEGY (%)

	Precision↑	Recall↑	F ₁ ↑
Baseline [61]	59.64	83.32	69.52
Ours	84.17	89.76	86.88
w GT	88.98	90.64	89.80

that the ablation study on the subject tracking features *has no impact* on the camera registration and tracking, i.e., the results in the first four columns.

Effectiveness of subject alignment feedback supervision: To validate the effectiveness of our proposed subject alignment feedback supervision strategy in Section III-C, we conduct experiments on BEVT-S as a multi-view human association task, which evaluates the model's ability to associate the same person across different views. We employ precision, recall, and F₁ score as evaluation metrics. As shown in Table III, we compare three settings: (1) the baseline model using ResNet-50 pre-trained in [61], (2) our proposed feedback supervision approach, and (3) the ResNet-50 trained with ground-truth association supervision (denoted as "w GT"). The results demonstrate that our feedback supervision strategy significantly improves the association performance, achieving an F₁ score of 86.88% compared to the baseline's 69.52%. This comprehensive improvement indicates more accurate and reliable person association across views. Moreover, our self-supervised approach achieves competitive performance compared to the ground truth supervised setting (89.80% F₁ score) with a very small gap, which further validates the effectiveness of the proposed self-supervised feedback supervision strategy.

D. Cross-Domain Evaluation on Real-World Dataset

We conduct cross-domain experiments where we train our method on the synthetic dataset BEVT-S and perform a cross-domain evaluation on the real dataset CVMHAT-BEV.

Geometric similarity-based localization metric: Since the absence of a scale between the real BEV and generated BEV,

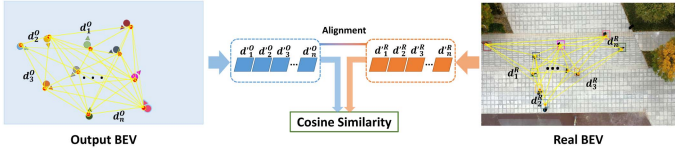


Fig. 9. An illustration of the geometric similarity-based metric. We label distances in output and real BEV as d_n^O and d_n^R respectively, and the normalized distances as $d_n^{O'}$ and $d_n^{R'}$. Before calculating similarity, these corresponding normalized distances will be aligned by IDs.

TABLE IV
REGISTRATION RESULTS FOR DIFFERENT NUMBERS OF VIEWS ON
CVMHAT-BEVT (%)

Methods	2 Views	3 Views	4 Views
BEVSee [19]	82.50	85.07	86.31
BEVSee + camera tracking	82.71	86.44	87.90
Ours	88.82	88.93	89.05

it is hard to directly evaluate the results on the real datasets. To address this problem, we propose a geometric similarity-based metric to evaluate the cross-domain performance of our method in detection and tracking problems. For a pair of real BEV (provided by the dataset) and generated BEV, we compute the distances among all subjects within each view to form a matrix. After normalization, we then flatten these two matrices into vectors by aligned IDs. We calculate the cosine similarity between these two vectors and use it to measure the detection result, as in Fig. 9.

Evaluation of the subject registration results: The CVMHAT-BEVT dataset contains a different number of side views. The proposed method has the generalization ability for the multi-view (≥ 3) scenarios. As discussed in Section III-F, we have employed the hierarchical maximum spanning sub-graph algorithm, which is specially used for multi-view subject fusion. We evaluated the geometric similarity metric in the multiple views (ranging from 2 to 4) scenarios in Table IV.

The results show that the geometric similarity between the BEV generated by our method and the real BEV is close to 90%. This indicates a strong cross-domain generalization of our approach and demonstrates its effectiveness in handling multiple views. Also, we can see that, compared to our preliminary work BEVSee, the proposed method obtains an obvious improvement. The proposed camera tracking strategy applied to the BEVSee can partly improve the registration performance. From the results, we can also find that the proposed method is not very sensitive to the number of views. The overall performance even slightly gets better with the increase of the number of views.

Evaluation of the subject tracking results: For the tracking task, we manually annotate the camera direction in the BEV of the real dataset. We then rotate the real and generated BEV to align with the same direction. After that, we normalize the positions of the subjects in both the real and generated BEV. Based on it, we use Metric III to evaluate the results.

As shown in Table V, we can see that, even though we use a cross-domain model trained on synthetic BEVT-S, the

TABLE V
COMPARISON OF CROSS-DATASET TRACKING RESULTS ON CVMHAT-BEVT (%)

	HOTA	MOTA	IDF ₁
BEVSee [19] + subject tracking	46.57	64.87	58.15
BEVSee + camera&subject tracking	51.55	66.42	60.76
Ours w/o camera tracking	53.26	67.31	61.91
Ours	57.59	70.10	64.91

TABLE VI
CROSS-DOMAIN GENERALIZATION ANALYSIS UNDER CHALLENGING
CONDITIONS ON CVMHAT-BEVT (%)

Condition	HOTA	MOTA	IDF ₁
Original	57.59	70.10	64.91
Heavy occlusions	55.31 (-2.3)	66.32	63.45
Crowd dynamics (1.5x)	56.99 (-0.6)	67.26	63.29
Crowd dynamics (2x)	53.51 (-3.2)	63.03	62.23

tracking results on the real dataset are still acceptable. We also compare our method to the preliminary version BEVSee [19]. Note that, BEVSee is an image-based method, which can not handle the tracking task directly. This way, we combine BEVSee with the proposed camera/subject tracking strategy to handle this problem. We can first see that the proposed camera and subject tracking strategy can be well adapted to BEVSee, and provides the acceptable performance. We can further see that, the combined method performs worse than our method. This demonstrates the advantage of the new model trained with the self-supervised spatial-temporal consistency learning loss.

Cross-domain generalization analysis: To validate the method's generalization to diverse real-world conditions, we conduct comprehensive experiments on challenging scenarios including complex lighting, heavy occlusions, and crowd dynamics. These experiments validate the robustness of our approach across different environmental conditions.

① *Complex lighting conditions:* To further demonstrate the robustness of our method under complex lighting conditions, we provide comprehensive visualization results in Fig. 11, where besides using the data in CVMHAT-BEVT, we also include several videos in [53] with various lighting conditions. Fig. 11 shows different lighting conditions that our method encounters, including (a) strong backlight glare, (b) shadow occlusion, (c) intense direct illumination (at noon), and (d) low-angle oblique lighting (at dusk). The figure presents both the real BEV ground truth and the corresponding tracking results generated by our algorithm. As can be observed, our method maintains excellent tracking performance even under these challenging lighting conditions, demonstrating strong robustness to illumination variations.

② *Heavy occlusion scenarios:* Following the definition of heavy occlusion in CrowdHuman [71], we define scenarios with horizontal target IOU > 0.7 as heavy occlusion cases. We conducted occlusion robustness tests on such video segments, which include a total of 2,651 frames across 6 video. As shown in the second row of Table VI, while there is some performance

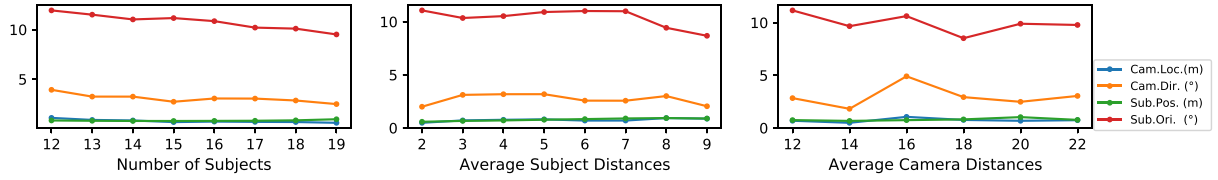


Fig. 10. Sensitive analysis of the proposed method regarding the number of pedestrians in the scene, and the distances among the locations of pedestrians/cameras.

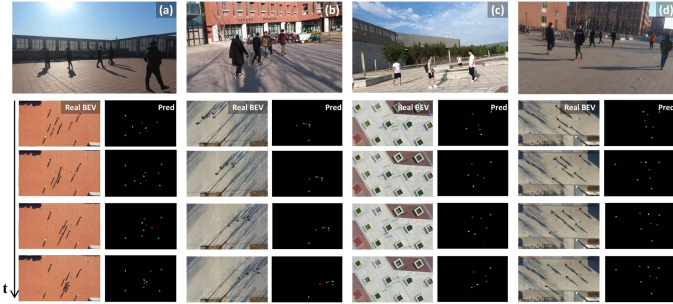


Fig. 11. Visualization of tracking results under various complex lighting conditions: (a) strong backlight glare, (b) shadow occlusion, (c) intense direct illumination, and (d) low-angle oblique lighting. The figure shows both the real BEV ground truth and the corresponding algorithm-generated tracking results, demonstrating robust performance across different lighting scenarios.

degradation, the overall decrease is only 2.3%, demonstrating the robustness of our method in handling occlusion scenarios.

③ *Crowd dynamics*: We simulate higher-intensity crowd dynamic scenarios by accelerating the original test videos by 1.5x and 2x speed. As shown in the last two rows of Table VI, at 1.5x speed, the algorithm performance remains essentially stable (HOTA decreases by only 0.6%), while at 2x speed, although there is performance degradation (HOTA decreases by 3.2%), the overall decline remains within reasonable limits. This demonstrates that our method possesses good temporal modeling capabilities and can effectively handle rapid changes in crowd movement patterns.

E. In-Depth Analysis

Pose estimation robustness under challenging scenarios: To address concerns about the method's dependence on accurate human pose estimation, we conduct systematic experiments on BEVT-S dataset to evaluate performance degradation under challenging conditions including low-resolution inputs and occluded views.

① *Low-resolution scenarios*: We downsample the original resolution (2704×1520) data to 1/2 (1352×760) and 1/4 (676×380) to simulate the impact of low-resolution inputs on pose estimation and overall tracking performance. As shown in Table VII, 2x downsampling causes almost no performance loss, while 4x downsampling results in only slight decreases in all metrics (e.g., localization error increases by only 0.04, HOTA decreases by 1.6%), demonstrating strong robustness to resolution variations.

TABLE VII
POSE ESTIMATION ROBUSTNESS ANALYSIS UNDER CHALLENGING SCENARIOS ON BEVT-S DATASET

Scenario	Cam.Loc.	Cam.Dir.	Sub.Pos.	Sub.Ori.	HOTA
Original	0.77	3.04	0.84	10.94	78.13
1/2 Res.	0.86	3.38	0.85	11.15	77.38
1/4 Res.	1.03	3.86	0.88	11.84	76.55
10% Occ.	0.79	3.12	0.85	11.14	78.08
20% Occ.	1.27	5.03	0.92	12.59	75.02

② *Occluded view scenarios*: We simulate extreme occlusion conditions by randomly adding 10% and 20% masks to subject regions, and then apply the pose estimation algorithm. As shown in Table VII, 10% occlusion has almost no impact on performance, while 20% occlusion results in only limited decreases in all metrics. This indicates that the complementary nature of multi-view information effectively mitigates the impact of single-view occlusion on overall tracking performance.

In summary, although our method relies on human pose estimation as initial input, the overall performance degradation under low-resolution and heavy occlusion challenging scenarios is very limited, validating the robustness and practicality of our approach.

Annotation economy and error analysis: Our self-supervised design significantly reduces annotation burden by requiring only camera relative positions as weak supervision information, eliminating the need for more difficult-to-annotate camera orientations or target calibrations. This represents a highly economical choice for practical applications. Furthermore, our method show good synthetic-to-real cross-domain generalization ability, which is fully validated by the experimental results. This means that it relies solely on synthetic datasets for training and can directly perform cross-domain inference on real data. Therefore, in actual deployment, the proposed method can be applied without any real data collection and annotation. Even when a small quantity of annotation for real data fine tuning is required, the necessary camera relative positions can be easily obtained through simple measurements (such as using a ruler and a protractor) without additional expensive equipment or complex operations.

In practice, we know that the simple measurements may involve a certain degree of error. This way, to better validate the robustness of our method under measurement errors, we designed two sets of experiments by randomly adding noise to the annotations of the BEVT-S dataset to address the robustness issues caused by camera distance and angle measurement errors.

TABLE VIII
ANGLE AND DISTANCE MEASUREMENT ERROR ROBUSTNESS ANALYSIS OF
CAMERA LOCATIONS

Condition	Cam.Loc.	Cam.Dir.	Sub.Pos.	Sub.Ori.	HOTA
Original	0.77	3.04	0.84	10.94	78.13
Angle error	1.32	4.12	0.89	12.34	76.92
Distance error	0.94	3.53	0.92	11.21	77.23

TABLE IX
TIME EFFICIENCY OF DIFFERENT COMPONENTS IN OUR METHOD

Module	VTM	Assoc.	Align.	Reg.	Track.	All
FPS	123	49	1,492	3,513	287	30

In the first set of experiments, we demonstrate the method's robustness to angle measurement errors. We randomly added angle deviations in the range of $[-15^\circ, 15^\circ]$ to the camera labels while keeping the distance between the two cameras accurate. The results are shown in Table VIII. In the second set of experiments, we demonstrate the method's robustness to distance errors. We randomly added distance errors in the range of $[-0.5 \text{ m}, 0.5 \text{ m}]$ to the camera labels while keeping the angles of the two cameras accurate. The results are shown in Table VIII. From the experimental results, we can see that given an error within one meter, the overall impact on results is very small. Even when facing angle errors as large as 15° , the method's performance changes remain minimal, especially for the comprehensive tracking metric HOTA, which shows very little impact. This effectively demonstrates the robustness of our method. In summary, our method demonstrates significant economic and practical advantages in both annotation costs and actual operations.

Sensitivity analysis: We provide the sensitivity analysis of our method for different scenes. Specifically, we investigate the impact of the number of pedestrians in the scene, and the distances among the locations of pedestrians or cameras. As shown in Fig. 10, we can see that the camera direction and subject orientation prediction results are more sensitive, but the overall fluctuation of the angle prediction stays basically within 2 degrees, while the position prediction results are quite stable within a very small range.

Latency analysis: Our method is an online multi-view tracking framework. Here we provide the latency analysis of the proposed method to evaluate its computational efficiency. As shown in Table IX, we compute the time efficiency of our method by measuring the average speed (FPS) of different modules and the overall speed. We can first see that the overall speed is *approximately real-time efficiency*. Moreover, the feature extraction operations in VTM and Association modules (Assoc.) in SAM take up most of the time, which can be implemented in parallel across multi-view inputs to further acceleration. The other modules including the multi-view BEV alignment (Align.), the registration and tracking models in our method are quite efficient. Our method achieves approximately 30 FPS on average, which means that for videos with frame rates of 30 FPS or below, there is virtually no latency in the system. This

TABLE X
ROBUSTNESS ANALYSIS AGAINST FRAME SYNCHRONIZATION DRIFT ON
BEVT-S DATASET

Frame Offset	Cam.Loc.	Cam.Dir.	Sub.Pos.	Sub.Ori.	HOTA
0	0.77	3.04	0.84	10.94	78.13
5	0.79	3.88	0.85	11.15	77.40
10	0.99	4.27	0.88	11.84	77.17
20	1.35	5.30	0.90	12.06	76.33
30	1.81	7.45	0.92	12.59	73.29

real-time performance makes our method suitable for online applications where immediate processing is required.

Frame synchronization analysis: Frame synchronization is crucial for multi-view tracking systems to ensure temporal consistency across different camera perspectives. In our implementation, we address this challenge through two approaches: (1) For the synthetic dataset (BEVT-S), we utilize Unity 3D's built-in frame synchronization capabilities to ensure perfect temporal alignment across all camera views. (2) For the real-world dataset (CVMHAT-BEVT), we employ manual timestamp alignment to synchronize frames from different cameras.

To validate the robustness of our method against synchronization drift, we conduct experiments with intentional frame offsets ranging from 0 to 30 frames. As shown in Table X, our method demonstrates strong robustness to synchronization errors. Specifically, with frame offsets up to 10 frames, the performance degradation is minimal, with camera localization errors remaining under 1 m and tracking metrics showing only slight decreases. Even with 20-frame offsets, the system maintains reasonable performance, with more significant degradation only occurring at 30-frame offsets or higher. This robustness is particularly important for real-world deployments where perfect synchronization may not always be achievable.

F. Qualitative Analysis

Fig. 12 shows two cases of the qualitative results. Specifically, we provide the generated BEV (Pred) from two first-person views on three frames, and the corresponding ground-truth BEV (GT). We can see that the prediction of both camera and subject registration can achieve a good coincidence with the ground truth, thanks to the effectiveness of our method for BEV registration and tracking. Note that, benefiting from the controllability of the synthesized data, we can also generate the real BEV by simulating placing a camera from a top view, as shown in the last column. We can see that, the predicted BEV can well reflect the relative position layout, face orientation and temporal trajectories of all the subjects in the real BEV (except an absolute in-plane rotation), which demonstrates the effectiveness of our method and the potential of BEVTrack problem. As shown in cases 1 and 2, we can also see that the proposed method can handle the various scenes, such as mutual occlusions, illumination variation, view difference, etc.

From the above qualitative results, we find that the proposed problem is very promising. It only requires the multi-view videos as input, and generates the *unified BEV map with the human position and face orientation, as well as the camera localization*

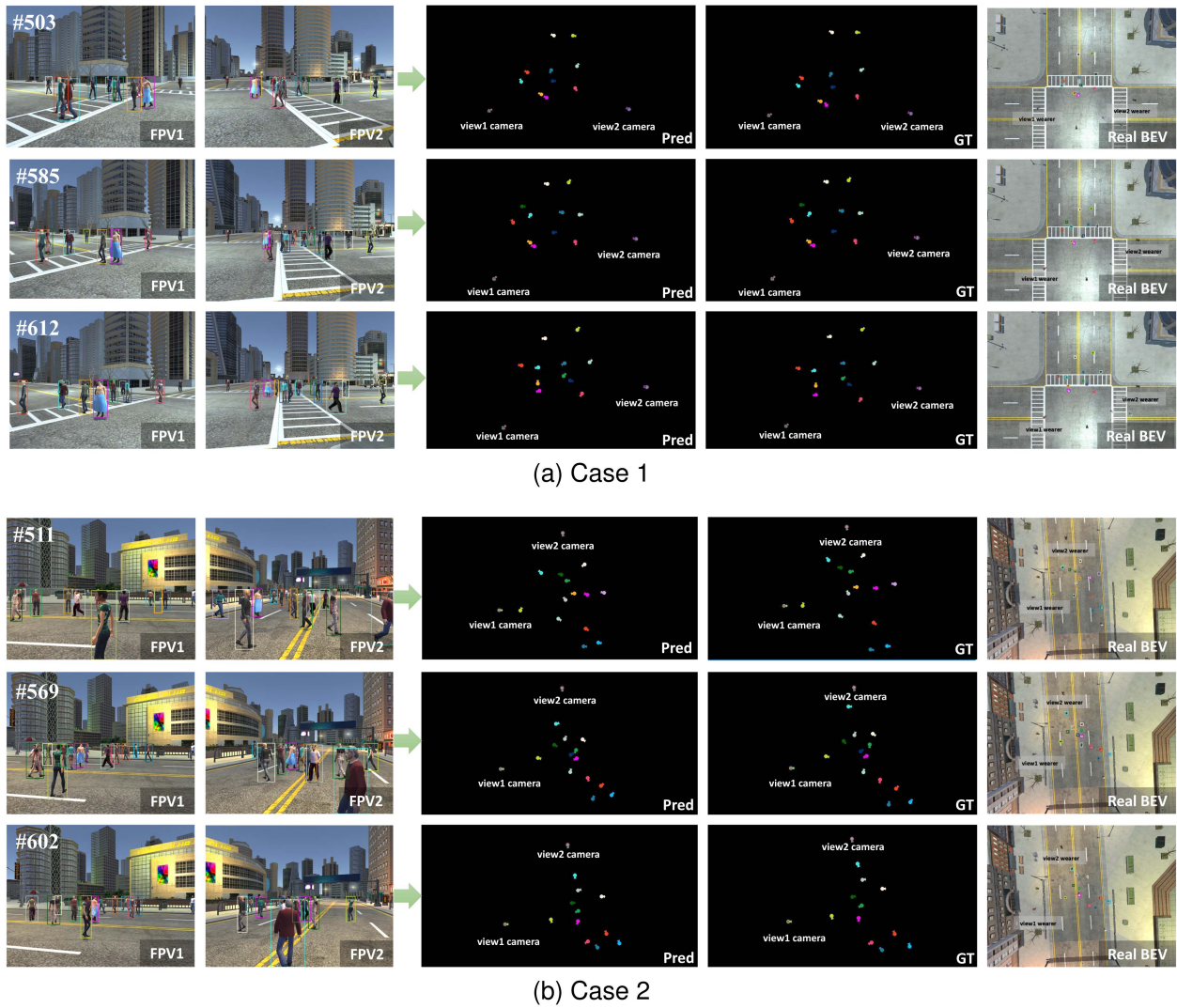


Fig. 12. Qualitative case analysis. Given the first-person views on each frame, we provide the generated BEV (Pred) and the corresponding ground-truth BEV (GT). We also simulate the real BEV videos for better illustration.

and field-of-view, along the continuous video. This significantly reduces the limitations of input and enriches the output results.

VI. DISCUSSION

Potential applications: The scenarios of using multi-view mobile cameras are very common in life. It is particularly suitable for scenes where it is difficult to pre-set the calibrated cameras, such as outdoor activities and the scenes with calibration changes, such as traffic surveillance cameras. For specific applications, our method can be applied to the video surveillance of some outdoor scenes, e.g., special person detection [51], interaction/group activity recognition and analysis [72]. However, it is hard to apply the calibration-based method used for fixed cameras. In this case, our method is useful, which only needs the multi-view videos as input and provides the BEV tracking with a global picture of the scene and clear trajectories for all subjects.

Limitation and future work: First, the proposed problem and method only focus on the subjects of pedestrians. In the future, we hope to expand our methods that can be applied to any object, not only pedestrians. We plan to integrate open-ended frameworks with text classification capabilities, such as VL-SAM [73], [74], to address this limitation and enable open-category object perception and tracking capabilities.

Second, our current method relies solely on RGB modality. Future work could explore integrating additional sensing modalities, such as LiDAR and Radar, to enhance robustness and accuracy. For instance, LiDAR depth cues can improve BEV perception as demonstrated in BEVFusion [35], while radar-camera BEV fusion methods like RCBEVDet [36] and RCBEVDet++ [37] show strong potential for complementary robustness in low-visibility or adverse weather conditions. By combining point cloud positions with subject poses, we can obtain more accurate subject center positions and orientations, leading to improved BEV multi-view registration and tracking results.

For the lack of a sufficient amount of annotated real datasets, we use the synthetic dataset for training and conduct cross-domain evaluations on real datasets. In the future, we plan to collect a large-scale real dataset for training and evaluation.

VII. CONCLUSION

In this work, we have proposed a new problem of BEVTrack, which aims to achieve the multi-view multi-target tracking in the BEV but only using the multiple FPV cameras without given calibration. To handle this problem, we propose a virtual BEV transform and alignment approach to generate a unified BEV from multiple side views. After that, we register and track both the camera and all the subjects in a complementary manner. We also create a new dataset with synthetic and real data for this problem. Experimental results show the effectiveness of our methods. Through the above effort, we hope to provide more research foundations for this challenging and valuable problem in the community.

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